MSG No. 93.121 $P4'_222'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `222.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `222.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,7,8]`

Table 5: Wyckoff site: `2e`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,4,7,8]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,3,5,6]`

Table 6: Wyckoff site: `2f`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,5,6]`

Table 7: Wyckoff site: `4g`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4]`
2	$[0,\ 0,\ -z]$	`[2,3]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8]`

Table 8: Wyckoff site: `4h`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8]`

Table 9: Wyckoff site: `4i`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[0,\ \frac{1}{2},\ -z\right]$	`[2,3]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8]`

Table 10: Wyckoff site: `4j`, site symmetry: `.2.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2]`
2	$[-x,\ 0,\ 0]$	`[3,4]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,7]`
4	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,8]`

Table 11: Wyckoff site: `4k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ x,\ 0\right]$	`[5,7]`
4	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[6,8]`

Table 12: Wyckoff site: `4l`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,2]`
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3,4]`
3	$[0,\ x,\ 0]$	`[5,7]`

continued ...

Table 12

No.	position	mapping
4	$[0,\ -x,\ 0]$	[6,8]

Table 13: Wyckoff site: 4m, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,2]
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3,4]
3	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[5,7]
4	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[6,8]

Table 14: Wyckoff site: 4n, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,7]
2	$\left[x,\ -x,\ \frac{3}{4}\right]$	[2,6]
3	$\left[-x,\ x,\ \frac{3}{4}\right]$	[3,5]
4	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[4,8]

Table 15: Wyckoff site: 4o, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{3}{4}\right]$	[1,7]
2	$\left[x,\ -x,\ \frac{1}{4}\right]$	[2,6]
3	$\left[-x,\ x,\ \frac{1}{4}\right]$	[3,5]
4	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[4,8]

Table 16: Wyckoff site: 8p, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[5]
6	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[6]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[8]